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## Intellectual Ventures

It was March 2009 and Nathan Myhrvold, co-founder and chief executive officer of Intellectual Ventures ("IV"), looked up at the wildlife photographs on his wall as he prepared for his next meeting. Myhrvold was often referred to as the former chief technology officer at Microsoft, but he was also known for his artistic photography, gourmet cooking and collection of dinosaur relics. In addition to his many accomplishments to date, Myhrvold hoped that within several years he would be recognized as the founder of "invention capital" – a new business model built around buying, developing, and monetizing intellectual property ("IP").

Myhrvold started Intellectual Ventures in 2000 with his former Microsoft colleague, Edward Jung. Their goal was to build a business that acquired and developed IP, and they planned to generate returns for their investors by selling or licensing the IP from the firm's portfolio. Myhrvold and Jung opened a small office near Seattle, Washington and began seeking capital from large technology companies and institutional investors. It took several years to secure the initial investments, but the founders had raised close to \$5 billion by the end of 2008, and they were still attracting new capital. As of September 2010, IV's portfolio topped 30,000 patents, a key metric for the firm.

Myhrvold and Jung were pleased with IV's progress in acquiring and developing a valuable patent portfolio that had generated approximately \$1 billion in cumulative licensing revenues. At the same time, the founders knew there were elements of the firm's business model that still needed to be refined. For example, although the company had licensing revenue, there was still perceived resistance to the model that had to be overcome. Also, the IV team was finding that some companies were reluctant to pay licensing fees, even though these companies were infringing upon IV's intellectual property.

The IV executives hoped to avoid suing companies in order to earn license fees, as litigious non-practicing entities (NPEs) had a bad reputation in the industry. Such firms were often referred to as "patent trolls," and IV did not want to be lumped into this category. Nevertheless, IV executives were ready and willing to file suit against an infringing company, if they saw no alternatives.

Another concern was the scalability of IV's primary means of generating revenue. The firm found that it took an average of 18 months of detailed discussions to execute a licensing arrangement, and the discussions required skilled attorneys and engineers that were familiar with industry-specific technologies. IV was eager to find a way to make revenue generation less labor intensive and costly. Another issue related to IV's ongoing ability to acquire the right patents at the right price in order to

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achieve attractive returns. As the intellectual property industry gained more visibility, the number of patents in the market was rising, as were the prices. Like others in the industry, the IV team found it difficult to evaluate the quality and value of each patent in a consistent and cost-effective manner.

Myhrvold thought about the firm's early successes as well as its challenges. IV had achieved its initial goals on fundraising and patent acquisition, and it was time to define the firm's next set of milestones. He was eager to engage his executive team in a discussion about the pros and cons of litigating for the license fees the firm believed it was owed. He knew that it could take years to win a lawsuit, but in the meantime, IV might need to signal that it was willing to fight. Myhrvold had warned his investors that they would need to be patient, and he knew the next several years would be a critical time to demonstrate the attractiveness of IV's business model.

## Overview of IP industry

The term "intellectual property" referred to creations of the mind such as inventions, literary, musical or artistic works, computer software, and symbols that could be protected through patents, copyrights, trademarks, and trade secrets. IP typically had commercial use, yet it was an intangible asset which made it difficult to assign value. Owners of intellectual property could choose to use, license and/or sell their IP. Licensing contracts were exclusive or non-exclusive, and typically specified terms such as duration, up-front payments, ongoing royalties, sub-licensing to other parties, modification rights and field of use (e.g., geographic restrictions). Some high tech companies (such as Qualcomm) generated nearly 40% of their revenue through licensing fees. Each country had its own IP regulations and policies; those of the U.S. are outlined below.<sup>1</sup>

### *Types of Intellectual Property*

**Patents** were agreements between inventors and the government in which an inventor provided a detailed description of an invention in exchange for the legal right to exclude others from making, using or selling the invention for 20 years. Patents were registered with the U.S. Patent and Trademark Office (USPTO), and upon expiration of the patent, the invention entered the public domain. All patent applications included *claims* that defined what was being protected. Types of claims included *product claims* covering the structure or composition of a product, *method claims*, which covered the methodology for producing the invention, and *product-by-process* claims in which the process for making a product was defined (e.g., in the chemical manufacturing industry). All patent claims had to satisfy the following criteria to be considered valid: new, useful, non-obvious and fully-disclosed. It could take several years for a new patent application to be approved by the USPTO, during which time the office researched other patents and inventions in the same "space." It typically cost an inventor between \$5,000 and \$25,000 to write and submit a patent application, and an inventor could choose to hire attorneys or other agents to advocate on their behalf before the USPTO. After a patent was issued, the patent holder needed to pay renewal and maintenance fees of \$930 after 3.5 years, \$2,360 after 7.5 years and \$3,910 after 11.5 years to maintain the patent.<sup>2</sup> Some experts believed that approximately half of all patents were abandoned before the full term of 20 years.

After a patent was issued, the patent holder was responsible for defending the patent against various forms of infringement. If a company or individual infringed upon a patent unknowingly, they were still held responsible and could be forced to pay license fees or damages. In some cases, plaintiffs sought injunctions against defendants, hoping to halt business activity that involved the IP in question. The U.S. Supreme Court ruled in a 2006 case (*eBay Inc. v. MercExchange L.L.C.*), that a plaintiff should not be granted an automatic injunction; instead, it depended on the facts in each case. This ruling was perceived as a win for manufacturing companies that feared the possibility of highly

disruptive injunctions. Some patent holders found it prohibitively expensive to pay the legal fees required to defend their patents, particularly against large companies with experienced IP attorneys on staff. (See **Exhibit 1** for patent litigation statistics.)

**Copyrights** protected authors and artists from unlicensed duplication and distribution of their creative expressions (e.g., architectural, audiovisual, literary, motion pictures, pictorial, graphic, sculptural, and sound recordings). To obtain copyright protection, the work had to be original, non-trivial, and recorded in a tangible medium. Unlike patents, copyrights were secured automatically when the work was created, and endured for the author's life plus another 70 years after the author's death. Reproducing, adapting, distributing, or publicly performing protected work without the copyright owner's permission constituted copyright infringement, and could lead to monetary damages that would be incurred by the infringer.

**Trademarks** were names, words, symbols, designs or slogans that distinctly identified the owner of the items marked. Under U.S. law, trademarks did not expire as long as the item was being used in commerce. From a legal standpoint, the more distinctive the trademark, the easier it was to protect. Trademark infringement consisted of the unauthorized use or imitation of a protected mark, and an infringing party could be forced to pay damages.

**Trade Secrets** were a means of protecting confidential items such as price lists, prototypes, drawings and formulas. They were protected through internal security measures such as marking items confidential or keeping documents hidden. Liability was incurred if a party acquired trade secrets through "improper means" such as theft, bribery or misrepresentation, or if parties entered into a confidential relationship, and then one party disclosed or used the trade secrets in an unauthorized manner.

### *Industry Trends*

By 2008, intellectual property was becoming an increasingly important factor in the world economy. The number of patent applications in the U.S. rose from 176,264 in 1990 to 484,955 in 2007, and the number of patents granted in the U.S. rose from 99,077 to 182,901. Almost half of the patents granted in the U.S. in 2007 were of foreign origin<sup>3</sup>. (See **Exhibit 2** for patent data from 1972-2007.) Companies — particularly those in the information technology sector — dramatically increased their IP activity during this period. For example, in 1990 Microsoft received five patents, but by 2005 the company was filing over 3,000 patent applications each year.<sup>4</sup> The use of copyrights, trademarks and trade secrets also increased during this time.<sup>5</sup> By some measures, this increase correlated with a shift in the overall U.S. economy from tangible to intangible assets. In 1975, more than 80% of the market value for the S&P 500 companies was composed of tangible assets such as factories, machines and inventory. By 2005, many of these companies had sold off or shut down their plants and outsourced manufacturing. At the same time, invention-based industries such as software, communications and medical devices were becoming more prevalent. As a result, tangible assets dropped to only 20% of market value, with intangible assets, particularly IP, making up the difference.<sup>6</sup>

Although intellectual property was growing as an asset class, the market for IP was still considered small and illiquid. The primary means of generating revenues from IP was through licensing deals executed by patent attorneys. By the mid 2000s, many technology companies started hiring dedicated IP professionals who were responsible for monetizing patents and negotiating license agreements with other IP owners.

## Industry Participants

### *Non-practicing entities*

There were over 325 active non-practicing entities (NPEs) in the U.S., most of which owned a few hundred patents.<sup>7</sup> The primary business model for these firms was acquiring patents and asserting the corresponding intellectual property rights by seeking licensing revenues from companies, often through litigation or the threat of litigation. The median price paid by NPEs for a patent was approximately \$100,000 and the mean \$400,000.<sup>8</sup> According to Stanford Law School's Intellectual Property Clearinghouse, there had been on average 2,500 patent infringement lawsuits every year from 2004-2007.<sup>9</sup> In 2008, the number was 2,896,<sup>10</sup> out of which approximately 500 had been brought by NPEs. The 2006 settlement in which Research-in-Motion (maker of the Blackberry smartphone) paid \$612.5 million to NTP, a Virginia-based patent-holding company, represented one of the most prominent examples of an NPE winning a major case. NPEs were also pejoratively known as "patent trolls." The phrase had initially been intended to refer to entities which held patents of dubious value, did not "practice" them (i.e., did not build products or services based on those patents), yet asserted them broadly against many players in the relevant industry, seeking settlements for large amounts of money and often asking for injunctions along the way.

### *Large technology firms*

Most of the biggest IP portfolios were managed by large technology firms such as IBM and Hewlett Packard. For example, IBM had over 26,000 active patents in the U.S. and over 40,000 worldwide. In 2007, IBM was granted 3,125 U.S. patents, more than any other company in the world. IBM's portfolio focused on six primary categories: microelectronics, servers, display, storage, network and e-business computing and software. IBM licensed a portion of its own portfolio to third parties, generating hundreds of millions of dollars of licensing fees on an annual basis. Other large technology firms managed deep IP portfolios in one or two sectors such as Qualcomm (cellular telephony) and Intel (semiconductors). Based on a recent patent survey conducted by the Institute of Electrical and Electronics Engineers (IEEE) the companies with the most "patent power" in the U.S. in 2008 were Microsoft, Intel and IBM. (See **Exhibit 3** for selected data from IEEE's patent survey and definition of patent power.)

Companies with large IP portfolios often entered into cross-licensing agreements with each other, covering part, or all, of a company's patent portfolio. In addition, large patent holders participated in patent pools in support of specific technology standards. Companies in these pools often entered cross-license agreements with other contributors to the pools.

### *Individual Inventors, Small Firms, and Universities*

A large fraction of U.S. inventions came from individual inventors or employees at small firms and universities. In fact, one statistic indicated that 43% of all patent applications in the U.S. in the 1990s came from this group.<sup>11</sup> These inventors faced challenges, however, with respect to monetizing their IP. They had a difficult time gaining enough leverage, credibility and financial resources to be taken seriously by large technology firms.<sup>12</sup> In addition, the inventors had a difficult time finding a company to build or license their inventions, and they rarely had the financial capacity to protect their IP from potential infringers. Furthermore, small inventors were often shut out of patent pools and other IP consortiums.

Inventions developed at universities typically belonged to the university; however, each university had its own system to manage and monetize the IP developed at that school. Many of the

larger and well-funded universities such as MIT and Harvard had active technology licensing offices that helped inventors establish license agreements or spin-offs, with the university keeping a portion of revenues or profits generated by the IP. (See **Exhibit 4** for 2008 statistics from the MIT Technology Licensing Office.) Most large universities had hundreds, or even thousands, of registered patents that were not being used or monetized in any material way. Many smaller universities did not get involved in licensing IP, so employees at those schools tended to pursue their own channels if they wanted to monetize inventions.

## Intellectual Ventures Background

The seeds of IV were sown while Myhrvold worked at Microsoft from 1986-1999. (See **Exhibit 5** for management bios.) Before serving as the chief technology officer at the company, Myhrvold had started a group called Microsoft Research that was dedicated to basic and applied research in computer science and software engineering. After leaving Microsoft to pursue his many hobbies, Myhrvold continued to think about the challenge of managing research and development within a large organization. He commented, “I concluded that IP was not run like a business. Someone gave a bunch of nerds some money and didn’t expect anything back.” He attributed this in part to the difficulty in generating and measuring returns from R&D efforts. In 2000, he teamed up with Jung, chief architect at Microsoft, to come up with a business that would address this opportunity. The founders looked to venture capital and private equity as potentially analogous businesses, and they hoped that invention capital would ultimately generate a similar level of profits. Myhrvold explained, “If you look back 30 years, few would have predicted that venture capital and private equity returns would have grown by a factor of 80. Initially, people had a difficult time investing in businesses that didn’t exist yet – the opportunities were seen as illiquid and ill-served by capital markets. The same can be said for inventions today. They are sources of great value, but not treated that way – we can help change that and create an enormous business.”

Myhrvold and Jung noted, however, that the success rate in identifying intellectual property winners was lower than that in venture capital. For example, while top venture firms expected to make money on 1 in 10 investments, IV hoped that 1 in 20 or 1 in 50 patents would turn out to be valuable. As a result, Myhrvold and Jung insisted from the beginning that they needed a portfolio of sufficient size to be successful. Myhrvold elaborated, “Any individual invention is very risky. It is one of the reasons that corporate research and development gets a bad name. But, I argue that the portfolio effect decreases the risk. The only way to make our model work is to have a large enough portfolio of unrelated inventions. Alternatively, if a problem is big enough, we can invest in multiple related inventions.” Jung expanded, “One of the key takeaways from my years at Microsoft is that it helps to invest in a platform and then be agnostic about what goes onto the platform. Companies and technologies fail for many different reasons that are hard to predict. It is better to have a portfolio covering multiple approaches than bet on just one.”

### *Funds*

Myhrvold and Jung’s emphasis on scale impacted the firm’s fund raising strategy. Their vision was to start with three core strategic investors that were not direct competitors. They believed they needed a minimum of \$1 billion of assets under management to be successful, and it was necessary to find patient investors that could wait five or seven years to see any results. Jung explained, “Some potential investors wanted to see complex financial projections for our firm, and we had to tell them that we would not be able to create meaningful projections for years.” It was reported that the key, early strategic investors included Microsoft, Nokia and Sony and that they were later joined by eBay, Google, Intel, Nvidia and SAP.<sup>13</sup> In exchange for providing the initial capital for the firm, the early

strategic investors were promised a broad set of licenses that covered current patents as well as future patents that would be acquired by IV in the future.<sup>14</sup> Different investors received different rights and, in general, investors/licensees who joined later received more expensive deals relative to early investors/licensees. (There were also licensees who were not investors and investors who were not licensees.) After securing capital from the initial strategic investors, IV pitched its invention capital model to institutional investors as well. IV was successful in raising money from institutional accounts such as pension funds and endowments.

By early 2009, IV had raised over \$5 billion across its four funds. The first and oldest fund was launched in 2003 and named Invention Science Fund I (ISF I). It focused on providing seed capital for building inventions “from scratch” (this included financing IV’s own invention laboratory). The second and third funds, Invention Investment Fund I and II (IIF I and IIF II), were launched in 2003 and 2008 respectively. They focused on acquiring existing (patented) inventions from all possible sources of invention - individual inventors to Fortune 10 companies.<sup>15</sup> The two IIF funds were the largest of the four funds. The fourth fund, Invention Development Fund I (IDF I), was launched in 2007 and focused on acquiring pre-filing inventions (i.e., inventions which have not yet been patented), mostly from universities in Asia, through a variety of technology transfer deals.<sup>16</sup>

For each of its funds, IV charged a management fee proportional to the value of assets under managements and “carried interest” proportional to the fund’s realized returns. This compensation structure was similar to that of most venture capital firms, private equity firms and hedge funds (which typically charged 2% in management fees and 20% in carried interest).<sup>17</sup>

### *IV Organization*

After Myhrvold and Jung committed to starting the firm, they invited two well-respected attorneys, Greg Gorder and Peter Detkin, to join them as co-founders in 2000 and 2002, respectively. The co-founders recruited slowly at first, hand-picking a few key hires such as Detkin’s former Intel colleague, Donald Merino, to join as general manager of acquisitions. Once IV started purchasing patents however, the management team realized they would need significantly more people to help with the technical analysis, due diligence and contract negotiations. The founders also decided to hire staff scientists. In 2010, the founders brought on board Adriane M. Brown as president and chief operating officer of IV.

By 2010, IV had grown to more than 700 people, most of whom were based in Bellevue, Washington, near Seattle. The others worked out of IV’s offices in Mountain View, Austin, Chicago, Australia, Dublin, and at IV’s five Asian offices (Korea, India, Singapore, China and Japan). The employees were split into three major functions: engineering, management/administrative, and legal. Jung expressed surprise with how many different types of jobs were needed to carry out the firm’s strategy. “This is a new industry without an existing ecosystem, and we have to do many things internally.”

## **Business Model**

### *IP Acquisition*

IV acquired its intellectual property through three primary methods, each of which corresponded to the focus of its four funds: buying (through IIF I & II), building (through ISF I) and partnering (through IDF I).

**Buying** - In the firm's early years, IV sought out patents by setting up meetings with companies and individual inventors and negotiating directly with IP owners (principals). Initially, Merino was concerned that IV would not find enough patents to buy. In fact, the firm purchased fewer than 100 patents in 2004. Merino explained, "When I joined IV, I asked myself, 'How do we turn this illiquid non-market into a real market?'" Merino increased IV's access to patents by establishing a network of "finders" and brokers who helped IV source patents from around the world in exchange for a commission on any consummated transaction. The better finders and brokers performed some due diligence before presenting the patents to IV, and Merino and his team were able to increase their purchase activity significantly by 2006.

IV started building its portfolio by concentrating on IP in sectors that would be valuable to its early investors such as software, communications and semiconductors. Over time, IV expanded into new segments, opportunistically purchasing portfolios that might prove valuable over time. Merino explained, "We have generally sought a balance between buying what we know is licensable right away versus more speculative investments. We seek the right combination in order to be attractive to licensees." IV sought depth in certain fields by purchasing multiple ways of solving a defined problem. Detkin explained, "In some areas, we need at least 10-12 patents in order to have something of value." Jung expanded, "We are fortunate that we don't suffer from the 'innovator's dilemma' – we will pursue all technology opportunities in a field without fear of cannibalizing ourselves."

Merino and his team analyzed each potential purchase by looking at the risks associated with the technology, product, and pricing elements to each deal. IV worked with its own set of engineers to assess the IP on a variety of measures: Did it offer substantial improvements to other technology? Was it going to be used in many kinds of products? In terms of product risks, Merino's team tried to determine if the IP was an essential patent, or is there a viable workaround? IV also looked at whether the IP was core or peripheral to a particular product or service.

Merino believed the pricing risks were often the hardest to assess given the lack of transparency in the industry. Although there were many firms specializing in IP, there was no broad aggregator of information. As Jung asserted, "Lots of firms are willing to price IP, but we find most prices are meaningless unless someone is willing to back up the prices with real dollars." Value was also dependent on each licensee's alternatives to the IP in question, and the level of exclusivity impacted pricing as well. Merino explained, "In the early days, pricing was pretty basic. We started with \$40,000 per patent as our baseline and went up or down from there. Now we have more transaction history from our own deals and those published by IP intermediaries such as Ocean Tomo. We are starting to create better guidelines that help us set prices." Pricing also fluctuated with overall market conditions. Merino noted that asking prices started to rise two to four times during the last quarter of 2008, which he attributed to the influx of "middlemen" and the availability of information about larger deals. (See **Exhibit 6** for Merino's internal pricing analysis.) Nevertheless, many individual inventors saw the value of selling to IV. As one inventor described, "I was planning to market my patents myself, but I paused to look at IV's offer. Could I have gotten more money? Very possibly. But how much work would it have taken? That was part of my decision [to sell to IV] – opportunity cost. This way I can put my time into my current start-up activities and work on other patents."<sup>18</sup>

Each potential acquisition could take weeks to analyze and execute. IV even had to create novel financial products to enable some of its acquisitions. One such product was the IP Financing Bridge. This was a deal structure, under which IV would provide capital to the acquiring firm in a merger, in exchange for taking ownership of the target firm's patent portfolio. The acquiring firm would then be granted a perpetual, non-exclusive license to those patents (cf. **Exhibit 7**). IV used this product in early 2009 to acquire 140 patents from Transmeta, a semiconductor company, after months of due diligence.<sup>19</sup> Another product created by IV was IP to Earnings-Per-Share (IP2EPS). This product was

designed for R&D-intensive companies, which were willing to grant IV full control over a subset of their patents, in exchange for obtaining more predictable revenue streams. This was achieved by transferring the relevant IP assets to a new subsidiary, wholly owned by the company, but unrestricted. The parent company would receive a free grant-back license to use those IP assets, whereas IV would acquire the exclusive right to sub-license the IP assets to anyone it chose, in exchange for fixed guaranteed payments, a share of licensing profits and IV taking over the patent prosecution and maintenance costs (cf. **Exhibit 8**). IV used IP2EPS in February 2009 to acquire the right to license 500 U.S. and international patents owned by Telcordia Technologies, a provider of telecommunications software and service solutions.<sup>20</sup>

Detkin explained, “There is no way to automate the process of IP valuation – we need to read and understand the patent claims.” Merino added, “The industry has changed a great deal in five years. We used to be concerned about finding enough patents to buy – now we are concerned that we don’t have the capacity to review all the patents that come our way. There has been a 90% increase in the number of patents shown to us in the last year and it still takes us almost as much time to review each patent. We are now working on filters to reduce the volume of patents that we review.”

**Building** - Intellectual Ventures also developed its own inventions through its team of scientists, technologists, business leaders, strategists, engineers, mathematicians, programmers, attorneys, IP experts and support staff. As of 2008, the company had signed up more than 100 senior inventors who worked with IV on a part-time basis: these included MIT professors and senior scientists from California’s Lawrence Livermore National Labs. (See **Exhibit 9** for a partial list of IV inventors.) Myhrvold was known for hosting events called “invention sessions,” during which an interdisciplinary group brainstormed about topics in a wide range of areas such as software, infectious diseases, energy resources, and advanced particle physics. The inventions developed during these sessions were owned by IV. In 2008, IV opened up an “invention lab” co-located with IV’s offices near Seattle.

By 2009, the company had filed more than 2,000 patent applications in more than 30 technology areas, making IV one of the most active invention groups worldwide. One of IV’s inventions, a new type of nuclear reactor, was selected by MIT as one of the top ten emerging technologies in 2009. IV believed the invention was so promising that it set up a new company, TerraPower, to develop the technology. By September 2010, TerraPower had 24 employees working out of an office near the IV headquarters. (See **Exhibit 10** for information about TerraPower.) The IV team was also working on a new weapon against malaria that targeted mosquitoes using sophisticated lasers. In 2009, IV had “thousands” of ideas under consideration. As Myhrvold explained, “When we develop an invention, we expect it will be 5-10 years out before it is successful, and then we have a time window until 20 years to enjoy the exclusivity granted to us through the patent. Unfortunately, we don’t know which of our inventions will be successful, so we have to make lots and lots of bets. That is one of the ways in which we are like a venture capital firm.”

**Partnering:** IV’s third channel for acquiring IP involved partnering with global institutions such as universities and government groups. In 2008, IV created a university program that formalized a way for IV to develop the IP being created by university researchers in exchange for a share of any future proceeds generated by that IP. In connection with this program, IV distributed Requests for Inventions (“RFIs”) that sought solutions to challenging world problems. IV reviewed the responses to its RFIs and offered to provide “invention capital” to those that looked promising. The capital included support for writing and filing patents as well as future marketing and licensing work. IV believed that it was helping inventors who typically didn’t have the resources to develop their ideas, thereby fulfilling a market need. (See **Exhibit 11** for a slide that IV presented at a conference about IV’s value proposition for inventors.)

IV developed comprehensive collaboration partnerships with universities, particularly leading science institutions outside the U.S. In March 2009, IV and the Indian Institute of Technology Bombay ("IITB") signed an agreement outlining a relationship in which IV planned to license, patent and commercialize IITB inventions. IV hoped to announce similar relationships with institutions in China and Singapore during 2009.

### *Monetizing IP and Inventions*

The primary way that IV generated returns for its investors was by selling non-exclusive licenses to strategic investors and other corporate licensees. Myhrvold explained, "When a VC or PE firm invests in a company, they are hoping to have a successful exit by going public or selling it to another business. These are options for us, but it is more likely that we will license patents to generate revenues. The equivalent in VC would be for a firm to hold on to a company and get a dividend from profits." IV executives believed that companies would want to enter into license agreements with IV for both defensive and offensive purposes. As Jung articulated, "IP transactions are fundamentally about risk management – companies want to defend against litigation or get a jump on new technologies." From a defensive standpoint, IV owned so many patents that most large technology manufacturing companies found it prudent to review IV's portfolio. Gorder elaborated, "The fact of the matter is that a single company rarely owns all of the invention rights for a complex product, and it's complicated for a company to find and license all of the IP that goes into a product."

IV believed that it provided a service by assembling a package of invention rights that could be licensed as a unit. Detkin asserted, "We are providing an efficient way for net users to pay net contributors. IV can provide the service of finding and acquiring all of the essential patents in a field and then selling non-exclusive licenses to multiple parties. Nevertheless, some companies see us as a nuisance and try to pay us as little as possible." Microsoft Corporation, the only company that publicly acknowledged its relationship with IV, believed IV played an important role in the IP ecosystem. Kenneth Lustig, managing director of IP acquisitions and investments at Microsoft, commented, "A key benefit for member companies is that IV has the resources to develop or obtain IP in areas where a member company has less of an invention focus. Because of the breakneck pace of our industry, and the volume and speed of new products coming to market, no company can do go it alone when it comes to innovation. Ultimately, our relationship with IV gives us more freedom to innovate, which then allows us to deliver better products to our customers."

Although many companies recognized that IV owned valuable patents, it was still difficult for IV to structure licensing deals that worked for both parties. According to Detkin, negotiations typically revolved around three key dimensions: the scope of the license (i.e., how many patents were included); the time horizon; and the price. Most potential licensees wanted to structure a deal with a one-time lump sum fee that covered all current and future patents, while IV sought shorter term deals limited to a sub-section of IV's portfolio in which the licensees paid on a running royalty basis similar to a subscription model. These divergent interests led to long, time-intensive and uncertain negotiations. Merino explained, "We find that every company wants a customized IP deal that involves detailed due diligence on our portfolio as well as intelligent bundling on our part. These deals can take 18 months or longer to negotiate, and they require experts to understand the nuances." As of early 2009, IV had close to 20 license agreements in place, primarily with companies that were also investors in one or more of the IV funds. It was reported in the media, for instance, that IV had negotiated investment and license deals ranging between \$200 million and \$400 million with Verizon Communications and Cisco Systems.<sup>21</sup> Detkin explained, "The number of customers could be higher, but we are focused on driving a high ROI for our investors so we only do deals that make economic

sense for us. We believe that it will become easier to do deals when we have a critical mass of licensees in each industry.” IV’s goal for the future was to sign up hundreds or even thousands of smaller companies for smaller amounts (\$5 million - \$10 million), using standardized licensing deals, as opposed to the current, highly customized deals.

IV was finding that it had greater success signing up licensees in industries that had a track record of paying for IP, such as software and communications. IV focused on these industries during its first few years in terms of attracting investors, purchasing assets, and negotiating individual deals. The IV executives tended to have the deepest personal connections in these industries as well, making it easier to get traction in the early stages of a deal. Merino and his team were starting to reach out to other industries, a process that required close coordination between the team purchasing assets and the group responsible for licensing deals. Merino noted, “We ask ourselves, ‘Do we determine the right technologies and then present them, or do we let the market inform our purchasing?’” IV also found it challenging to pursue license deals in industries with many small companies such as e-commerce, as it was not time-efficient to approach each of the companies. Kevin Barhydt, director of acquisitions, explained, “It takes almost the same amount of time to do a small deal as it does to do a big deal.” IV was working on a method of executing small license agreements in a cost-efficient way, potentially by helping to create an e-commerce industry license association. The firm was also exploring shorter and less expensive “standardized” licenses with smaller companies in order to gain scale.

As of August 2009, IV had spent \$315 million buying patents from individual inventors, \$135 million on its invention labs, and \$848 million acquiring IP assets from SMEs under various deal structures (e.g., IP financing bridge and IP2EPS). In the same time frame, IV had returned approximately \$1 billion in patent royalty revenue to its investors.<sup>22</sup> Myhrvold described these numbers as “strong interim results.” IV executives were increasing their focus on revenue generation, and the firm was in the process of assembling six different industry teams to develop non-exclusive licensing packages for companies in each industry. These teams would also work with the acquisition group to suggest additional IP that would strengthen IV’s offering in that sector. Under Brown’s leadership, IV wanted each team to generate \$300-\$400 million per year of revenue in the next few years, which represented a big step up from the current revenue run rate.

IV was also open to offering exclusive licenses in industries such as health-care that required extremely high levels of investment to get a product to the market. For example, Pfizer spent billions developing Lipitor, but it has resulted in over \$10 billion in sales. In addition, IV hoped to make money over time by starting successful companies based on the IP developed in-house such as TerraPower.

## Conclusion

As Nathan Myhrvold thought about the opportunities and challenges that lay ahead, he was excited with the progress his firm had made so far – IV had the largest IP portfolio held by any non-manufacturing company, and had signed license agreements with some of the largest technology firms in the world. He was also pleased with the IP he and his colleagues had developed in-house. Myhrvold believed they had hired and partnered with some of the best inventors in the U.S. and abroad.

These early signs showed great potential, but Myhrvold and his team knew that they still had some hurdles to clear to generate the returns they and their investors wanted. The management team was thinking about how IV’s acquisition strategy impacted its ability to generate revenues. As Myhrvold explained, “In the past, people have not made money in IP because they got the time scale

wrong or the amount of money wrong. People said ‘let’s try it with \$30 million,’ but any 30 million dollars could be wrong one way or another. It’s like writing life insurance policies with too small a group of people. It’s all about scale.” Although the diversity enabled IV to reduce the risks associated with exposure to any particular industry or technology, it made it difficult for IV to monetize the portfolio in a cost-efficient manner. Myhrvold wondered how they should monitor the effectiveness of their acquisitions. Should they make adjustments to the firm’s screening and acquisition process?

The IV team also reflected on its methods of generating revenues. While the early licensees were paying IV for the IP they were using, other companies in the same sector continued to balk at the prospect of signing the types of license agreements that were acceptable to IV. Should IV be more flexible in the types of deals it was willing to sign? Could they create a standardized deal template to shorten the deal-making process? IV was also considering the use of litigation to force these companies to pay for what they were using. Detkin commented, “We believe that we can gain compliance through the strength of our portfolio, but we will sue companies as a last resort.” The IV executives wondered when it was appropriate to file the first lawsuit, and they were thinking carefully about which company or companies they should target.

Myhrvold thought about his business in light of the various constituents that played a role in IV’s ecosystem: investors, licensees, inventors and the IV team. As the IV team developed this new industry of invention capital, they had to balance the needs of each of these groups. There were bound to be some conflicts of interest that required careful management. Myhrvold recalled the original vision of IV as he reflected on the firm’s future, “If we can combine capitalism and invention, we can make the world a better place.” He wondered what else IV could be doing at this point to achieve that vision.

**Exhibit 1** Patent Litigation Statistics

- Legal fees and expenses for litigation through trial for cases in which \$500,000 to \$1,000,000 is at stake total about \$500,000 for each side.
- Legal fees and expenses for litigation through trial for cases in which \$1 million to \$10 million is at stake total about \$1.3 million for each side.
- The average time to resolve patent suits is 1.1 years.
- Seventy-six percent of patent suits settle, while 17% are transferred or thrown out on pre-trial motions.
- Patentees prevail 58% of the time at trial. Due to pre-trial losses, the patentee's chance for overall success in litigation is only about 49%.
- Patentees win 68% of the time before a jury if they file suit first, whereas they win only 38% of the time before a jury if the accused infringer files first.
- Judges throw out 15% of cases before trial, and roughly half of that 15% are thrown out for invalidity.
- At trial, patentees face a 33% chance that their patents will be invalidated.
- Forty-four percent of published decisions that validate a patent are appealed.
- Thirty-four percent of infringement-related verdicts are appealed and the federal circuit overturns 20% of them.

Source: Samson Vermont, "Risk and Reward: More Patent Facts and Stats, *Ideas to Assets: Investing Wisely in Intellectual Property* (New York: John Wiley & Sons, 2002), cited in Kellogg School of Management, Northwestern University technical note "Intellectual Property—The Ground Rules," (Boston: Harvard Business Publishing).

**Exhibit 2** Patent Data—1967–2007 (will update to 2008)

| Year | Total Patent Applications | Total Patent Grants | % of Total Grants that are of Foreign Origin |
|------|---------------------------|---------------------|----------------------------------------------|
| 1972 | 105,300                   | 78,185              | 30%                                          |
| 1977 | 108,377                   | 69,778              | 36%                                          |
| 1982 | 117,987                   | 63,276              | 40%                                          |
| 1987 | 139,455                   | 89,385              | 47%                                          |
| 1992 | 186,507                   | 107,394             | 45%                                          |
| 1997 | 232,424                   | 124,069             | 44%                                          |
| 1998 | 260,889                   | 163,144             | 44%                                          |
| 1999 | 288,811                   | 169,086             | 44%                                          |
| 2000 | 315,015                   | 175,979             | 45%                                          |
| 2001 | 345,732                   | 183,972             | 46%                                          |
| 2002 | 356,493                   | 184,376             | 47%                                          |
| 2003 | 366,043                   | 187,015             | 47%                                          |
| 2004 | 382,139                   | 181,302             | 48%                                          |
| 2005 | 417,508                   | 157,717             | 48%                                          |
| 2006 | 452,633                   | 196,404             | 48%                                          |
| 2007 | 484,955                   | 182,901             | 49%                                          |
| 2008 | 485,312                   | 185,224             | 50%                                          |

Source: USPTO, patent trademark and monitoring team, [http://www.uspto.gov/go/taf/us\\_stat.htm](http://www.uspto.gov/go/taf/us_stat.htm), accessed April 2, 2009.

**Exhibit 3** Selected Data from IEEE Spectrum Patent Scorecard (companies with adjusted pipeline power scores above 1,400)

| Company                        | Sector                   | 2007 US Patents | Pipeline Growth Index | Pipeline Impact | Self-Citations | Adjusted Pipeline Impact | Pipeline Generality | Pipeline Originality | Pipeline Power | Adjusted Pipeline Power |
|--------------------------------|--------------------------|-----------------|-----------------------|-----------------|----------------|--------------------------|---------------------|----------------------|----------------|-------------------------|
| Microsoft Corp (US)            | Software                 | 1649            | 1.12                  | 1.37            | 30%            | 1.37                     | 1.38                | 1.00                 | 3505           | 3505                    |
| Intel Corp (U.S.)              | Semiconductor            | 1865            | 0.95                  | 1.24            | 20%            | 1.24                     | 1.23                | 1.03                 | 2796           | 2796                    |
| IBM (U.S.)                     | Manufacturing            | 3149            | 0.86                  | 0.96            | 13%            | 0.96                     | 1.06                | 1.00                 | 2747           | 2747                    |
| Samsung Electronics (S. Korea) | Computer Systems         | 2749            | 1.11                  | 0.87            | 18%            | 0.87                     | 0.83                | 0.95                 | 2096           | 2096                    |
| Roche Holding AG (Switzerland) | Manufacturing            | 515             | 1.51                  | 1.44            | 19%            | 1.44                     | 1.65                | 1.03                 | 1902           | 1902                    |
| Hitachi (Japan)                | Biotechnology and Pharma | 2728            | 0.85                  | 0.87            | 23%            | 0.87                     | 0.98                | 0.95                 | 1874           | 1874                    |
| Micron Technology (U.S.)       | Electronics              | 1484            | 0.92                  | 1.25            | 64%            | 0.83                     | 1.38                | 1.09                 | 2582           | 1707                    |
| 3M Co (U.S.)                   | Semiconductor            | 471             | 0.97                  | 1.73            | 46%            | 1.45                     | 2.18                | 1.17                 | 2004           | 1674                    |
| Seiko Epson Corp (Japan)       | Manufacturing            | 1218            | 1.00                  | 1.06            | 25%            | 1.06                     | 1.09                | 1.01                 | 1416           | 1416                    |
|                                | Conglomerates            |                 |                       |                 |                |                          |                     |                      |                |                         |
|                                | Computer Peripherals     |                 |                       |                 |                |                          |                     |                      |                |                         |
|                                | and Storage              |                 |                       |                 |                |                          |                     |                      |                |                         |

Source: 1790 Analytics, shown on IEEE Spectrum, <http://www.spectrum.ieee.org/patentsurvey2008>, accessed April 2, 2009.

Notes from IEEE Spectrum: The number for 2007 U.S. patents is a proxy for relative patent prowess worldwide. The **Pipeline Power** score is derived by multiplying the company's patent count by the product of four other variables. **Pipeline Growth** (not shown here) represents the firm's 2007 patent activity, relative to its average performance in the five previous years. For the other three variables, a score above 1.00 indicates that the company performed better than average in its technology class; below 1.00 indicates worse than average performance. **Pipeline Impact** indicates how frequently all 2007 patents cited a company's patents from the previous five years. **Pipeline Generality** is a measure of the variety of technologies drawing on a company's patents. **Pipeline Originality** measures the variety of the technologies upon which an organization's patents build. **Adjusted Pipeline Impact** eliminates self-citation. The final score, **Adjusted Pipeline Power**, is an estimate of a company's overall patent power.

**Exhibit 4** Statistics from the MIT Technology Licensing Office for 2008

|                                                                                                    |          |
|----------------------------------------------------------------------------------------------------|----------|
| Total Number of Invention Disclosures .....                                                        | 522      |
| Number of U.S. Patents Filed (including provisionals, follow-ons, etc.) .....                      | 282      |
| Number of U.S. Patents Issued.....                                                                 | 122      |
| Number of Licenses Granted (not including trademarks and end-use software) .....                   | 68       |
| Number of Trademark Licenses Granted .....                                                         | 19       |
| Number of Software End-Use Licenses Granted .....                                                  | 15       |
| Number of Options Granted (not including options as part of research agreements) .....             | 30       |
| Number of Companies Started venture capitalized and/or with minimum of \$50K of other funding..... | 20       |
| Cash Income.....                                                                                   | \$89.1 M |
| Royalties .....                                                                                    | \$65.9 M |
| Patent Reimbursement .....                                                                         | \$8.2 M  |
| Equity Cash-In .....                                                                               | \$ 4.0 M |
| Expenditure on Patents.....                                                                        | \$16.3 M |

Source: TLO Statistics for Fiscal Year 2000–2009, [http://web.mit.edu/tlo/www/about/office\\_statistics.html](http://web.mit.edu/tlo/www/about/office_statistics.html), accessed April 2, 2009.

**Exhibit 5** IV Management Bios

**Nathan Myhrvold, Founder and CEO**—At Intellectual Ventures, Myhrvold is focused on a variety of business interests relating to the funding, creation and commercialization of inventions. During his 14-year tenure at Microsoft, Dr. Myhrvold held various positions within the company and was responsible for founding Microsoft Research and numerous technology groups that resulted in many of Microsoft's most successful products. In 1986, his company Dynamical Systems was acquired by Microsoft. Prior to that, he was a postdoctoral fellow in the department of applied mathematics and theoretical physics at Cambridge University and worked with Professor Stephen Hawking on research in cosmology, quantum field theory in curved space time and quantum theories of gravitation. Dr. Myhrvold holds 18 patents and has more than 100 patents pending. Dr. Myhrvold earned a doctorate in theoretical and mathematical physics and a master's degree in mathematical economics from Princeton University. In 2005, in recognition of his distinguished career, Princeton awarded Dr. Myhrvold the James Madison Medal, the university's top honor for alumni. He also has a master's degree in geophysics and space physics and a bachelor's degree in mathematics, both from UCLA.

**Edward Jung, Founder and President**—Before Intellectual Ventures, Jung was chief architect and advisor to executive staff at Microsoft Corporation. During his 10 years at Microsoft, Jung managed projects relating to web platforms, semantic web technology, intelligent operating systems, adaptive user interfaces and artificial intelligence. Jung co-founded many Microsoft teams including Windows NT, Microsoft Research, mobile and consumer products and web services. Before joining Microsoft in February 1990, Jung ran the Deep Thought Group working on neural network chips for learning and parallel computation. He also consulted to and wrote software for NeXT Computer, Apple Computer and its Advanced Technology Group, and the Open Software Foundation. An avid inventor, Jung holds more than 70 patents and has more than 500 patents pending.

**Gregory Gorder, Founder and Vice Chairman**—Prior to joining Intellectual Ventures in March 2000, Mr. Gorder spent 14 years providing business and legal advice to early stage technology companies as a lawyer for Perkins Coie LLP, the Northwest's leading law firm. While at Perkins Coie, Mr. Gorder helped dozens of companies with initial and follow on public offerings, acquisitions and divestitures, working closely with investment bankers, accountants and management teams. After graduation from law school, Mr. Gorder spent his first year clerking on the United States Court of Appeals for the Ninth Circuit for Judge Eugene Wright. Before starting law school, Mr. Gorder worked two years with Arthur Andersen & Co. as a software systems consultant. Mr. Gorder holds a B.A. in Business Administration from Washington State University and a J.D. from the University of Washington.

**Peter N. Detkin, Founder and Vice-Chairman**—Prior to joining Intellectual Ventures, Mr. Detkin spent eight years at Intel Corporation where he was a vice president and assistant general counsel. As assistant general counsel, Mr. Detkin was responsible for managing the Intel patent and licensing departments, including all aspects of prosecution and claims management. In addition, Mr. Detkin managed the litigation and competition policy departments (including antitrust). Before he joined Intel, Mr. Detkin was an intellectual property partner at the law firm of Wilson, Sonsini, Goodrich and Rosati in Palo Alto, Calif. Mr. Detkin was the first patent lawyer hired at this firm and formed the basis of their highly successful intellectual property practice. Mr. Detkin received his B.S.E.E. with honors in 1982 from the University of Pennsylvania's Moore School of Electrical Engineering, and a J.D. in 1985 from the University of Pennsylvania Law School.

**Don Merino, Senior Vice President, Channel Licensing**—Prior to joining Intellectual Ventures, Dr. Merino spent four years at Intel Corporation where he was director, strategic business development. In this position, Dr. Merino was responsible for managing the Intel licensing program. This program produced more than \$250M per year in revenue. Before Intel, Dr. Merino was director of licensing for General Instrument where he participated in the development of the MPEG Licensing Authority (MPEG-LA), an organization founded to license essential patents on a core standard-based technology. Additionally, he was responsible for the successful divestiture of General Instrument's fiber optics patent portfolio to Corvis Corp. for over \$230M in cash and equity. Upon graduation from the U.S. Naval Academy Dr. Merino served as an officer in the U.S. Navy first as an Engineering officer on the Battleship New Jersey (BB-62) and was also the navigator on the Cruiser Arkansas (CGN-41). He received a Navy Commendation Medal and a Navy Achievement Medal for his service. Dr. Merino received his B.S. from the U.S. Naval Academy in 1984 and his M.E.M.E and Ph.D. from Stevens Institute of Technology.

Source: Intellectual Ventures.

## Exhibit 6 IV Internal Pricing Analysis

Key Deal Metrics: \$/Deal from December 2006–December 2008

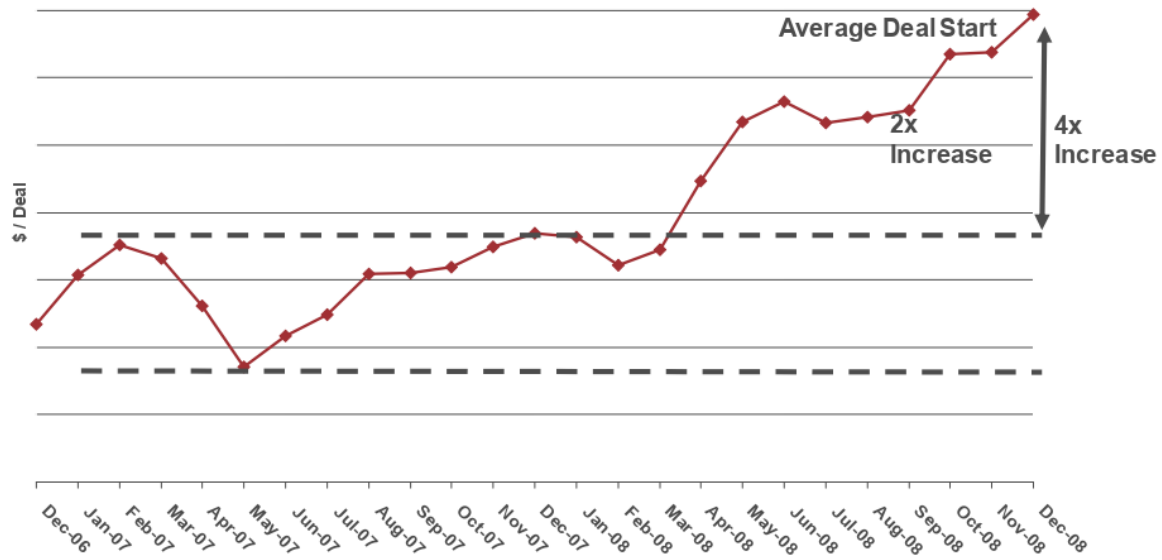
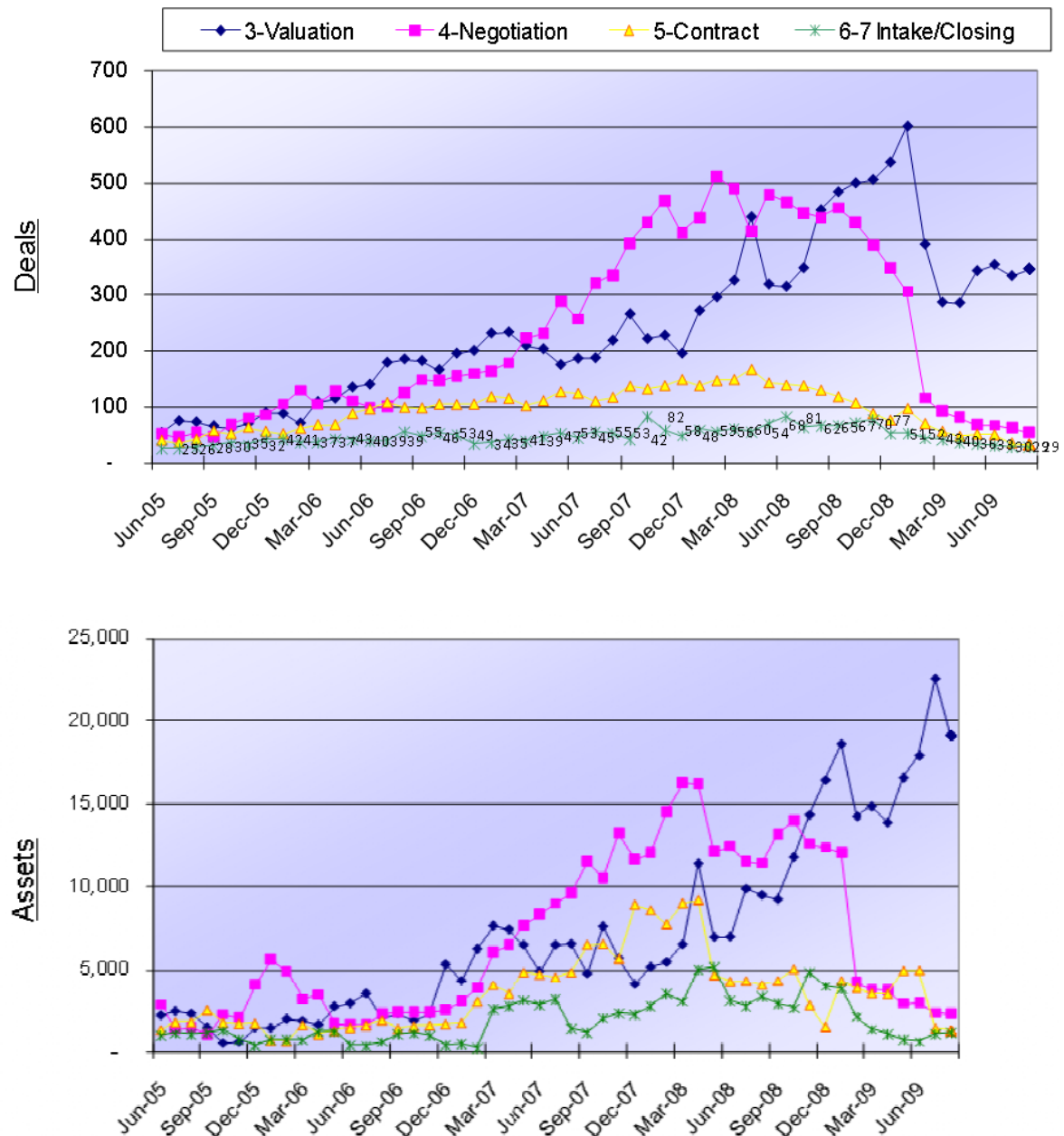


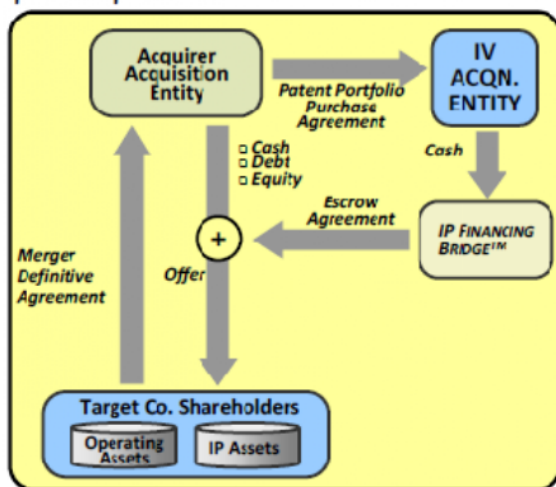
Exhibit 6 (continued) Deal Pipeline (June 2005–February 2009)



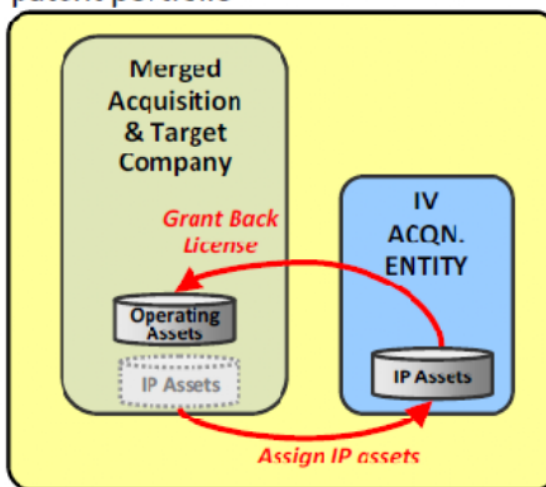
Source: Intellectual Ventures.

Exhibit 7 IP Financing Bridge™ for MM&amp;A Deals

**Pre-Closing:** IV contributes additional cash in escrow to bridge the offer in exchange of a pre-agreed sale of patent portfolio



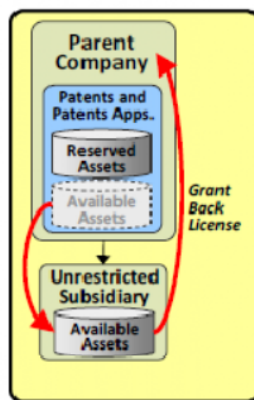
**Post Closing:** If offer is successful, buyer assigns the patents to IV and receives a grant back license to the patent portfolio



Source: Intellectual Ventures.

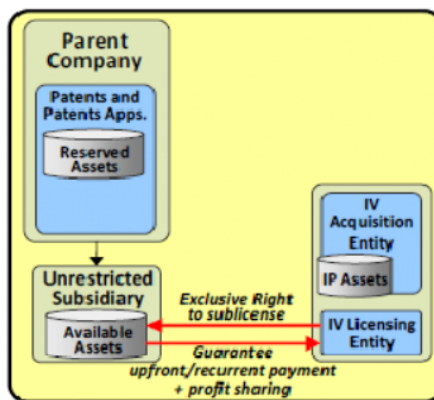
Exhibit 8 IP-to-EPS™

Assignment of Available IP assets to a wholly-owned unrestricted subsidiary



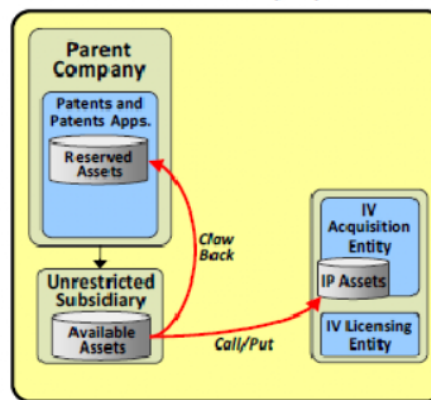
Sale of an exclusive right to sublicense the Available IP Assets in exchange for mix of:

- + upfront payment
- + recurring payments
- + profit sharing



**Call/Put options** enable to assign IP assets to IV and avoid entanglements between the licensing activities and the operating business

**Claw Back Rights** enable to obtain back full control on the non-assigned IP assets for defensive purpose



Source: Intellectual Ventures.

**Exhibit 9** Partial List of IV Inventors

**Edward Jung, Founder, President**—Edward Jung is a founder and President of Intellectual Ventures. Mr. Jung co-founded Intellectual Ventures after leaving Microsoft Corporation where he was chief architect and advisor to executive staff.

**Nathan Myhrvold, Founder, CEO**—Nathan Myhrvold founded Intellectual Ventures after retiring from his position as chief strategist and chief technology officer of Microsoft Corporation.

**Casey Tegreene, VP and Chief Patent Counsel**—Casey Tegreene is Vice President and Chief Patent Counsel at Intellectual Ventures where he focuses on a variety of projects relating to invention and innovation.

**Ken Caldeira, Staff Scientist, Carnegie Institution of Washington, Department of Global Ecology**—Ken Caldeira is a Staff Scientist at the Carnegie Institution, wherein his only job responsibility is “to make important scientific discoveries.”

**Alexander Cohen, Senior Analyst**—Alexander J. Cohen is a senior analyst at Quantum Intellectual Property Services, where he focuses on software patent analysis and portfolio analysis and management for clients.

**Bran Ferren, Co-Chairman, Applied Minds, Inc.**—Mr. Ferren is a designer and technologist working in entertainment, product development, engineering, architecture and the sciences.

**John S. Foster, Jr., Chairman of the Board of GKN Aerospace Transparency Systems**—Dr. John S. Foster, Jr. is Chairman of the Board of GKN Aerospace Transparency Systems, member of the board of Wackenhut Services, Inc., and consultant to Northrop Grumman Corp., Sikorsky Aircraft Corp., Titan Corp., Ninesigma, Technology Strategies & All

**W. Daniel Hillis, Co-Chairman and Chief Technology Officer, Applied Minds, Inc.**—Danny Hillis is co-chairman and chief technology officer of Applied Minds, Inc.

**Leroy Hood, Co-Founder and Director, Institute for Systems Biology**—Dr. Leroy Hood is recognized as one of the world’s leading scientists in molecular biotechnology and genomics and was a major contributor to the Human Genome Project.

**Roderick Hyde, Senior Scientist and Inventor**—Dr. Hyde recently joined Intellectual Ventures as a senior scientist and inventor. Previously he spent 32 years as a senior physicist at University of California’s Lawrence Livermore National Laboratory.

**Muriel Y. Ishikawa, Senior Scientist and Inventor**—During her 18 years at Lawrence Livermore National Laboratory, Dr. Ishikawa was a project and program leader of new-concept technical innovative projects.

**Alois A. Langer, Senior Inventor**—Born in Pittsburgh, Pennsylvania, Dr. Alois A. Langer graduated from the Massachusetts Institute of Technology in 1967 with a degree in electrical engineering and later from Carnegie Mellon University with a Ph.D. in electrical engineering/biotechnology.

**Robert Langer, Professor, Massachusetts Institute of Technology**—Robert S. Langer is one of 14 Institute Professors (the highest honor awarded to a faculty member) at the Massachusetts Institute of Technology (MIT).

**John Latham, Emeritus Professor, University of Manchester, UK**—John Latham's principal current research interests are in the fields of cloud physics, atmospheric electricity and global climate.

**Eric C. Leuthardt, M.D., Neurosurgeon, Washington University in St. Louis**—Eric C. Leuthardt, M.D. is a neurosurgeon who is currently an assistant professor with the Department of Neurological Surgery at Barnes Jewish Hospital and the Department of Biomedical Engineering at Washington University in St. Louis.

**Roy Levien, Managing Inventor, Ambient Architecture**—Roy Levien is Principal and Manager of Keystone Advantage, LLC, and a Managing Inventor at Ambient Architecture, an invention company.

**Mark Malamud, Principal and Manager, busymonster, LLC**—Mark Malamud is the principal and manager of busymonster, LLC, a consultancy company focused on advanced user interface and design.

**Bill Mangione-Smith, Senior Engineer and Inventor**—William Henry Mangione-Smith has been a professional engineer for over 15 years.

**Sir John Pendry, Professor, Imperial College London**—Professor Sir John Pendry FRS has been professor of Theoretical Solid State Physics at Imperial College London since 1981.

**John D. Rinaldo, Jr, User Interface Architect**—John Rinaldo has participated in several Intellectual Ventures' invention sessions relating to user interface.

**David Ruzic, Director of the Center for Plasma Material Interactions at the University of Illinois at Urbana-Champaign**—Dr. David N. Ruzic is the Director of the Center for Plasma Material Interactions at the University of Illinois at Urbana-Champaign.

**David R. Smith, Augustine Scholar and Professor, Duke University**—Dr. David Smith earned his Ph.D. in Physics from the University of California at San Diego, and then worked as a postdoc researcher on plasmon resonant particles as optical labels for biological and biochemical assays.

**Willard H. (Bill) Wattenburg, Research scientist at the Research Foundation, California State University, Chico; scientific consultant to the Lawrence Livermore National Laboratory**—Dr. Willard H. (Bill) Wattenburg is presently a research scientist at the Research Foundation, California State University, Chico, and a scientific consultant to the Lawrence Livermore National Laboratory where he conducts most of his experimental work.

**Thomas Weaver, Senior Interviewer, Fannie and John Hertz Foundation**—Dr. Thomas Weaver is senior interviewer for the Fannie and John Hertz Foundation.

**Chuck Whitmer, Software Architect**—Chuck Whitmer was co-founder of Dynamical Systems Research, Inc. with Dr. Nathan Myhrvold.

**Richard N. Zare, Blake Wilbur Professor, Stanford University**—Richard N. Zare is the Marguerite Blake Wilbur Professor in Natural Science at Stanford University. He was born on November 19, 1939 in Cleveland, Ohio, and is a graduate of Harvard University, where he received his B.A. degree in chemistry and physics.

Source: Intellectual Ventures.

## Exhibit 10 TerraPower Overview

# Introducing TerraPower

## The Need for Innovation in Energy

The world faces an urgent need for sources of electricity. Many initiatives seek options that are cleaner than fossil fuels, more reliable and independent of geography than wind and solar power, and safer and less expensive than the large international system that supports existing nuclear plants. Encouraging advances in technologies have been made in recent years. Science and engineering efforts have produced new ways to generate electricity, to use energy more efficiently, and to capture greenhouse gas emissions and prevent their entry into the atmosphere. Although the magnitude of the global energy challenge is enormous, nuclear energy can be an important part of the solution.

Intellectual Ventures™ investments in energy inventions have evolved to become TerraPower, an expert team that is investigating innovative ways to address energy needs. TerraPower's most advanced work centers on radically improving ways to make electricity using nuclear reactors.

TerraPower's innovative reactor designs could avoid the well-known limitations of current plants, nearly all of which employ nuclear reactor technology that dates back to the 1950s. Today's reactor designs require rare, enriched uranium fuel, but can extract only a small fraction of the energy in that fuel. As a result, these reactors produce too much waste, consume too much of our dwindling and increasingly costly uranium resources, and require frequent and expensive refueling.

The TerraPower initiative is advancing a comprehensive program to improve the technology of reliable nuclear plants. Its business approach presents an important alternative to create a financially and socially attractive emission-free energy that is safe and sustainable.

## TerraPower is Investing in Invention

### Invention Labs

Bringing together expert staff and individual consultants from some of the most prestigious nuclear laboratories and energy companies in the world

Public

Academic

Private



### Solution Set

Exploring important improvements to nuclear power by using openly shared information, state-of-the-art computational capabilities, and expanded data

### Invention Research & Development

Fine-tuning ideas for energy-producing reactions and finding new ways to apply the nuclear physics



### Market Enablement

Meeting industry demand for electricity and nuclear energy with better solutions to waste and proliferation concerns

## Physics Enables a Class of Nuclear Reactors

Led by John Gilleland, the TerraPower team relies on effective partnerships with leading experts in nuclear science, engineering, and industry who share a common vision. Gilleland's past experience with a variety of energy technologies supports TerraPower's multi-disciplinary approach to solving problems.

Based on recent advances in applied nuclear physics and technology made by pioneering American national laboratories and international academic groups, TerraPower recognizes that self-sustaining and self-limiting nuclear reactions are feasible in unenriched fuels. Extensive computer simulations and engineering studies underway at Intellectual Ventures have produced encouraging evidence that a wave of fission moving slowly through a fuel core could generate a billion watts of electricity continuously for well over 50 to 100 years without refueling. This new class of reactors, known as traveling-wave reactors (TWRs), could enable nations to simplify the nuclear

infrastructure dramatically. TWRs could reduce the amount of nuclear waste by using existing stockpiles of depleted uranium as fuel. By extracting centuries worth of energy from waste at enrichment plants, the TWRs would turn a social and financial liability into an asset.

## COMPARING REACTOR TECHNOLOGIES

| GENERATION III                                                                                                                                                                                                 | GENERATION IV                                                                                                                                                           | TRAVELING-WAVE REACTOR                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- Currently operating technology</li> <li>- Water-cooled thermal reactors</li> <li>- Once-through fuel cycle in the United States</li> <li>- 5% burn-up rate</li> </ul> | <ul style="list-style-type: none"> <li>- Fast breeder reactors</li> <li>- Closed fuel cycle</li> <li>- Multiple recycling processes used to increase burn-up</li> </ul> | <ul style="list-style-type: none"> <li>- Fast-neutron reactor that breeds its own fuel</li> <li>- Once-through fuel cycle</li> <li>- 20% burn-up rate</li> </ul> |

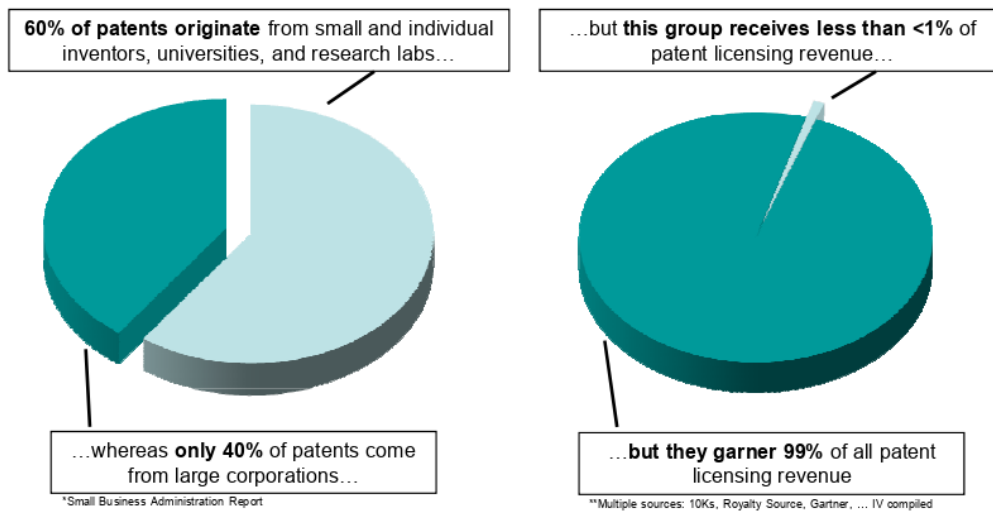
For press inquiries: [TerraPowerInfo@intven.com](mailto:TerraPowerInfo@intven.com)

INTELLECTUAL  
VENTURES

Source: Intellectual Ventures.

## Exhibit 11 IV Value Proposition for Inventors

## Patent Creation and Revenue Imbalance



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INTELLECTUAL  
VENTURES

Source: Intellectual Ventures.

## Endnotes

<sup>1</sup> The following article served as a reference for the overview section of this case: James G. Conley and David Orozcom, "Intellectual Property – The Ground Rules," Kellogg School of Management, Published by Harvard Business School Publishing, August 30, 2005.

<sup>2</sup> Fees based on FY2007 fee schedule published by USPTO. Available at [http://www.uspto.gov/web/offices/ac/qs/ope/fee2007september30\\_2007dec17.htm](http://www.uspto.gov/web/offices/ac/qs/ope/fee2007september30_2007dec17.htm), accessed May 12, 2008.

<sup>3</sup> Origin was defined based on the residence of the first named inventor listed.

<sup>4</sup> "Survey: A Market for Ideas," *The Economist*, October 22, 2005, page 4.

<sup>5</sup> Ibid.

<sup>6</sup> Ned Davis Research, as provided by Ocean Tomo.

<sup>7</sup> Patent Freedom, <https://www.patentfreedom.com/research.html>, accessed December 14, 2010.

<sup>8</sup> <https://www.patentfreedom.com/research-lot.html>, accessed December 14, 2010.

<sup>9</sup> Zusha Elinson, "Surprises in stats for IP suits: Stanford counters standard notions," *The Recorder*, December 10, 2008.

<sup>10</sup> "Acacia: The Companies Tech Loves to Hate," *Businessweek*, February 1, 2010. [http://www.businessweek.com/technology/content/feb2010/tc2010021\\_504247.htm](http://www.businessweek.com/technology/content/feb2010/tc2010021_504247.htm)

<sup>11</sup> Peter N. Detkin, "Leveling the Patent Playing Field," *John Marshall Review of Intellectual Property Law*, 2007, page 637.

<sup>12</sup> Peter N. Detkin, "Leveling the Patent Playing Field," *John Marshall Review of Intellectual Property Law*, 2007, page 641.

<sup>13</sup> Amol Sharma and Don Clark, "Tech Guru Riles the Industry By Seeking Huge Patent Fees," *The Wall Street Journal*, September 17, 2008.

<sup>14</sup> IV did not publicly disclose its investors, but names such as Microsoft, Nokia, Sony and Apple appeared in several articles about the firm.

<sup>15</sup> Nigel Page, "TV Shifts Gears," *Intellectual Asset Management Magazine*, Issue 36, July/August 2009.

<sup>16</sup> Nigel Page, "TV Shifts Gears," *Intellectual Asset Management Magazine*, Issue 36, July/August 2009.

<sup>17</sup> Nigel Page, "TV Shifts Gears," *Intellectual Asset Management Magazine*, Issue 36, July/August 2009.

<sup>18</sup> Intellectual Ventures Investor Profile: Robert Osann, Spring 2009.

<sup>19</sup> Nigel Page, "TV Shifts Gears," *Intellectual Asset Management Magazine*, Issue 36, July/August 2009.

<sup>20</sup> Nigel Page, "TV Shifts Gears," *Intellectual Asset Management Magazine*, Issue 36, July/August 2009.

<sup>21</sup> Amol Sharma and Don Clark, "Tech Guru Riles the Industry By Seeking Huge Patent Fees," *The Wall Street Journal*, September 17, 2008.

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